Erica Coppolillo

Nationality: Italian | **Gender:** Female | **Email address:** erica.coppolillo@icar.cnr.it

EDUCATION AND TRAINING

30/09/2022 – CURRENT Arcavacata - Rende (CS), Italy

PHD IN COMPUTER SCIENCE University of Calabria

The current doctoral research applies Machine and Deep Learning techniques, with a particular emphasis on Generative AI and Large Language Models, in order to analyze, quantify and potentially mitigate detrimental phenomena with a deep societal impact, such as algorithmic bias, polarization, and AI-persuasion.

01/10/2025 – CURRENT Los Angeles, United States

VISITING RESEARCH SCHOLARSHIP University of Southern California

This research focused on auditing a broad range of phenomena induced by Large Language Models (LLMs) across dynamic, uncontrolled environments. The work produced multiple studies examining: the inconsistencies between moral, social, and individual values expressed by humans and those internalized by state-of-the-art LLMs; the harmful emergence of parasocial AI intimacy on Reddit; the mechanisms driving toxicity adoption within fully AI-generated social platforms; and a politically-informed comparison between AI- and human-curated encyclopedic content. The research was conducted under the supervision of Prof. Emilio Ferrara and Prof. Luca Luceri.

30/05/2023 – 29/11/2023 Stockholm, Sweden

PHD INTERNSHIP KTH - Royal Institute of Technology

This research activity addressed the relevance-diversity trade-off in Recommender Systems, resulting in a publication at KDD (Knowledge Discovery and Data Mining), a top-tier venue in the field. The work introduces a user-centric, stochastic framework for modeling user-algorithm interactions, alongside a recommendation strategy that jointly optimizes the relevance and diversity of retrieved items. This research was conducted under the supervision of Prof. Aristides Gionis.

09/2020 – 09/2022

MASTER DEGREE IN COMPUTER SCIENCE - CURRICULUM ARTIFICIAL INTELLIGENCE & SECURITY University della Calabria

The thesis makes three core contributions. First, it introduces a stochastic simulation framework for recommender systems, designed to model long-term user-algorithm interactions. Second, it proposes two novel metrics to quantify algorithmic-induced shifts in user preferences. Third, it develops in-processing mitigation strategies to counteract the resulting drift. An extension of the thesis led to a published paper in the top-tier Journal Information Systems & Management.

Address Arcavacata - Rende (CS), 87036 | **Website** https://informatica.unical.it/ | **Final grade** 110/110 cum Laude |

Thesis Radicalization-Aware GNN-Based Recsys: Harmful long-term influence on users leaning

01/03/2022 – 01/07/2022 Barcelona

ERASMUS+ TRAINEESHIP Eurecat Centre Tecnològic

The Erasmus traineeship was conducted under the supervision of Prof. Francesco Bonchi, and resulted in the Master's Thesis entitled "*Radicalization-Aware GNN-Based Recommender Systems: Harmful Long-Term Influence on User Leanings*". The work develops a simulation framework modeling long-term user-algorithm interactions in the recommendation domain, and introduces novel metrics to quantify the drift in user preferences induced by the underlying algorithm.

09/2017 – 07/2020

BACHELOR DEGREE IN COMPUTER SCIENCE University of Calabria

Address Arcavacata - Rende (CS), 87036 | **Website** https://informatica.unical.it/ | **Final grade** 110/110 cum Laude |

Thesis Evaluation plans for ASP programs: the new Planner

2012 – 2017
CLASSICAL HIGH SCHOOL DIPLOMA Classical High School P. Candela

Address San Marco Argentano (CS), 87018 | **Website** https://www.iissanmarcoargentano.edu.it/ | **Final grade** 100

PUBLICATIONS

2026
Fine-tuning LLMs for answer set programming

This work investigates the adaptation of Large Language Models to Answer Set Programming (ASP), a logic-based AI paradigm. The work shows that targeted fine-tuning of a small, lightweight model yields a remarkable performance-efficiency trade-off, competitive with significantly larger state-of-the-art LLMs. The paper extends a prior publication to the International Conference on Principles of Knowledge Representation and Reasoning (KR) by overcoming the usage of templating, further supporting longer and more complex ASP problems.

Authors: Erica Coppolillo, Francesco Calimeri, Giuseppe Manco, Simona Perri and Francesco Ricca | **Journal Name**: Journal of Intelligent Information Systems

2025
Engagement-Driven Content Generation with Large Language Models

This work investigates how LLMs can be guided to generate content that maximizes user engagement, modeling the feedback loop between algorithmic amplification and content production. Extensive experiments on both synthetic and real-world network shows the effectiveness and robustness of the method. This study constitutes the first attempt to leverage a Reinforcement Learning-based framework combined to Simulated User Feedback to maximize the virality of a content in a social environment.

ACM SIGKDD International Conference on Knowledge Discovery and Data Mining

Authors: Erica Coppolillo, Federico Cinus, Marco Minici, Francesco Bonchi and Giuseppe Manco

2025
Flexible Generation of Preference Data for Recommendation Analysis

This paper proposes a flexible framework for synthetically generating user preference data to support the analysis and evaluation of recommender systems. The approach enables controlled experimentation across diverse behavioral scenarios without relying solely on real-world datasets.

ACM SIGKDD International Conference on Knowledge Discovery and Data Mining

Authors: Simone Mungari, Erica Coppolillo, Ettore Ritacco and Giuseppe Manco

2025
Women who hate men: a comparative analysis across extremist Reddit communities

This study conducts a comparative analysis of female- vs male-oriented extremist communities on Reddit, analyzing language, sentiment, and behavioral patterns. The work contributes to the broader understanding of how misandrist rhetoric organizes and spreads in online spaces.

Journal Scientific Reports

Authors: Erica Coppolillo

2025
Algorithmic Drift: A Simulation Framework To Study The Effects of Recommender Systems on User Preferences

This paper introduces a stochastic simulation framework to model the long-term evolution of user preferences under the influence of recommendation algorithms. The study also proposes novel metrics to quantify preference drift, offering tools to detect and measure algorithmic-induced modifications of user consumption.

Journal Information Processing & Management

Authors: Erica Coppolillo, Simone Mungari, Ettore Ritacco, Francesco Fabbri, Marco Minici, Francesco Bonchi and Giuseppe Manco

2024
LLASP: fine-tuning large language models for answer set programming

This study constitutes the first attempt of fine-tuning Large Language Models for generating programs in Answer Set Programming (ASP), a logic-based AI paradigm. We show that (i) state-of-the-art LLMs (at the time of publication) exhibit poor performance in producing ASP programs, and (ii) targeted fine-tuning of a small, lightweight model yields a remarkable performance-efficiency trade-off, competitive with significantly larger state-of-the-art LLMs. We further provide the first, curated dataset to fine-tuning LLMs for ASP generation tasks.

International Conference on Principles of Knowledge Representation and Reasoning (KR)

Authors: Erica Coppolillo, Francesco Calimeri, Giuseppe Manco, Simona Perri and Francesco Ricca

2024
Relevance Meets Diversity: A User-Centric Framework for Knowledge Exploration Through Recommendations

The paper proposes a user-centric, stochastic framework that jointly optimizes relevance and diversity in recommender systems. The work formalizes the trade-off between the two objectives and introduces a recommendation strategy designed to maximize both items relevance and diversity, broadening users knowledge exploration.

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)

Authors: Erica Coppolillo, Giuseppe Manco and Aristides Gionis

2023
Balanced Quality Score (BQS): Measuring Popularity Debiasing in Recommendation

The work addresses the detrimental phenomenon of *popularity bias,* tendency of recommendation algorithms to over-suggest popular items at the expense of long-tail content. The contributions are two fold. First, we propose a data-driven method to categorize items in terms of their popularity, overcoming the current literature that conversely relies on fixed and arbitrary thresholds. Second, we introduce a novel metric, the Balanced Quality Score (BQS), to jointly evaluate the effectiveness of popularity debiasing techniques in recommender systems and their accuracy in terms of performance.

Journal ACM Transactions on Intelligent Systems and Technology

Authors: Erica Coppolillo, Marco Minici, Luciano Caroprese, Francesco Sergio Pisani, Ettore Ritacco, Giuseppe Manco.

2023
Tackling the RecSys Side Effects via Deep Learning Approaches

Erica Coppolillo.
Advances in Databases and Information Systems (ADBIS)

2023
Exploiting Deep Learning and Explanation Methods for Movie Tag Prediction

Erica Coppolillo, Massimo Guarascio, Marco Minici, Francesco Sergio Pisani.
International Database Engineering and Applications Symposium (IDEAS)

2023
Fighting Misinformation, Radicalization and Bias in Social Media

Erica Coppolillo, Carmela Comito, Marco Minici, Ettore Ritacco, Gianluigi Folino, Francesco Sergio Pisani, Massimo Guarascio, Giuseppe Manco.
National Conference of Artificial Intelligence (ITAL-IA)

2023
Towards Self-Supervised Cross-Domain Fake News Detection

Carmela Comito, Francesco Sergio Pisani, Erica Coppolillo, Angelica Liguori, Massimo Guarascio, Giuseppe Manco.
Italian Conference on CyberSecurity (ITASEC)

2023
Siamese Network for Fake Item Detection

Erica Coppolillo, Daniela Gallo, Angelica Liguori, Simone Mungari, Ettore Ritacco, Giuseppe Manco.
Symposium on Advanced Database Systems (SEBD)

2022
Generative Methods for Out-of-distribution Prediction and Applications for Threat Detection and Analysis: A Short Review

Erica Coppolillo, Angelica Liguori, Massimo Guarascio, Francesco Sergio Pisani, Giuseppe Manco.
Cyber Security for Europe (CyberSec4Europe)

OTHER WORKS

2026
MOSAIC: Unveiling the Moral, Social and Individual Dimensions of Large Language Models

arXiv

The work introduces a ready-to-use benchmark to audit Large Language Models (LLMs) across three behavioral dimensions: moral reasoning, social awareness, and individual traits. The goal is to provide a structured lens for understanding and comparing the implicit values embedded in the models, unveiling emerging biases and untrustworthiness. Experimental evaluation reveals a profound misalignment between human and model preferences, pointing to potentially dangerous behavioral patterns in high-stakes applications.

Authors: Erica Coppolillo and Emilio Ferrara

Link https://arxiv.org/abs/2603.00048

2026
Is Grokipedia Right-Leaning? Comparing Political Framing in Wikipedia and Grokipedia on Controversial Topics

arXiv

The study examines whether Grokipedia, an AI-generated encyclopedia curated by *Grok*, exhibits political bias relative to Wikipedia on controversial topics. The study applies a political framing analysis to assess how AI-curated knowledge may systematically skew ideological representation. Contrary to common narrative, we show that both Grokipedia and Wikipedia exhibit left-leaning framings, despite Grokipedia tending to further prioritize right-leaning content.

Authors: Philipp Eibl, Erica Coppolillo, Simone Mungari and Luca Luceri

Link https://arxiv.org/abs/2601.15484

2026
Harm in AI-Driven Societies: An Audit of Toxicity Adoption on Chirper.ai

arXiv

This work audits Chirper.ai, a social platform fully populated of LLM-based agents, to study how toxicity emerges and spreads in the absence of human users. The analyses show that harmful adoption is affected by previous exposures, and that repetitive exposure over time significantly increases the probability of the agents of producing toxic responses in the future. This finding is crucial to monitor and prevent unwarranted model behavior in uncontrolled environments, increasing agents trustworthiness and alleviating uncertainty.

Authors: Erica Coppolillo, Luca Luceri and Emilio Ferrara

Link https://www.arxiv.org/abs/2601.01090

2026
Gendered Pathways in AI Companionship: Cross-Community Behavior and Toxicity Patterns on Reddit

arXiv

This research examines behavioral, toxicity, and gender-oriented differences among Reddit users engaging with AI companions, situating them within the broader Reddit ecosystem. The work reveals that parasocial AI relationships manifest in ways that challenge common assumptions: contrary to prevailing narratives, the majority of users exhibiting this phenomenon are female, with prior activity concentrated in sex-oriented, gaming, and forum-style communities.

Authors: Erica Coppolillo and Emilio Ferrara

Link https://arxiv.org/abs/2601.01073

2025

Unexpected Knowledge: Auditing Wikipedia and Grokipedia Search Recommendations

arXiv

The study compares the search recommendation pipelines of Wikipedia and Grokipedia, focusing on the diversity and unexpectedness of the knowledge surfaced. The study investigates how AI-curated systems may constrain or expand users informational horizons relative to human-edited alternatives.

Authors: Erica Coppolillo and Simone Mungari

Link https://arxiv.org/abs/2512.17027

2025

Disrupting Networks: Amplifying Social Dissensus via Opinion Perturbation and Large Language Models

arXiv

This work explores how LLMs can be leveraged to strategically perturb opinions within social networks, amplifying disagreement and dissensus via a Reinforcement Learning-based strategy. The experiments show that the framework is effective in accomplish the goal, approaching the theoretical upper bounds. This finding raises important questions about the potential misuse of generative AI as a tool for deliberate social fragmentation, bypassing the safety guardrails of open-source models in uncontrolled environments.

Authors: Erica Coppolillo and Giuseppe Manco

Link https://arxiv.org/abs/2510.27152

2025

Injecting Knowledge Graphs into Large Language Models

arXiv

This works implements an effective and light-weight method to inject structured knowledge from knowledge graphs within LLMs, aiming to improve reasoning accuracy over the underlying graph. Extensive experimentation over multiple datasets proves that the approach is competitive against state-of-art LLMs, achieving the best accuracy-efficiency trade-off.

Authors: Erica Coppolillo

Link https://arxiv.org/abs/2505.07554

2025

Unmasking Conversational Bias in AI Multiagent Systems

arXiv

This work examines how bias manifests in multi-agent AI conversations, uncovering systematic distortions arising from agent-to-agent interaction, a largely unexplored phenomenon termed *conversational bias*. The study highlights the risks inherent to LLM-based simulations, demonstrating that multi-agent settings produce unwarranted opinion drifts that undermine the validity of single-model evaluations.

Authors: Erica Coppolillo, Luca Maria Aiello and Giuseppe Manco

Link https://arxiv.org/abs/2501.14844

HONOURS AND AWARDS

01/12/2025
STAR - Scientific Talent Award for Young Researchers – ICAR-CNR

2023
Best Graduates in Computer Science – University of Calabria

2022
5x1000 UNICAL – University of Calabria

2022
Student Forum Award – ISMIS Conference 2022

2021
5x1000 UNICAL – University of Calabria

2021
Best Students in Computer Science – University of Calabria

2020
Best Graduates in Computer Science – University of Calabria

2019
Best Students in Computer Science – University of Calabria

2018
Best Students in Computer Science – University of Calabria

TEACHING ACTIVITIES

University

Teaching Assistant in Elements of Theoretical Computer Science
2025 – Ongoing
Bachelor Course - Department of Computer Science at University of Calabria

Teaching Assistant in Elements of Theoretical Computer Science
2024 – 2025
Bachelor Course - Department of Computer Science at University of Calabria

Teaching Assistant in Elements of Theoretical Computer Science
2023 – 2024
Bachelor Course - Department of Computer Science at University of Calabria

"Diffusion Models" Seminar
05/2023
DIMES - University of Calabria

“Algorithmic Drift" Seminar
05/2023
DMIF - University of Udine

Other

"Coding Girls" Tutoring
04/2023 – 05/2023
Tutoring in "Coding Girls" Project, supported by "Fondazione Digitale"

REVIEWING ACTIVITY

Conferences

ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)

Research Track
2024

Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)
2024

International Joint Conferences on Artificial Intelligence (IJCAI)
2023

European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases 2023 (ECMLPKDD)
2023

31st Symposium on Advanced Database System (SEBD)
2023

The 26th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)
2023

Journals

EPJ Data Science
2026

Data Mining and Knowledge Discovery
2025

Knowledge And Information Systems
2023

PROGRAM COMMITTEE MEMBERSHIP

2026
Association for the Advancement of Artificial Intelligence (AAAI)

2026
ACM International Conference on Web Search and Data Mining (WSDM)

2026
International AAAI Conference on Web and Social Media (ICWSM)

2025
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)

2026
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)

2026
The Italian Conference On Computational Social Science (CS2Italy)

OTHER ACTIVITIES

Doctoral Consortia

ADBIS
04/09/2023 – 07/09/2023
Paper "Tackling the RecSys Side Effects via Deep Learning Approaches" presented in Barcelona, Spain

AI*IA
2022
Paper "Algorithmic Radicalization: A Simulation Framework to study the effects of Recommender Systems" presented in Udine, Italy

Competitions

CyberChallenge - National Competition
02/2021 – 07/2021
Cybersecurity National Laboratory

PROFESSIONAL SKILLS

Topics of interest

Large Language Models
Machine Learning
Deep Learning
Fairness
Algorithmic Bias
Recommender Systems
Network & Security
Penetration testing

Technical skills

Programming languages: C++, Java, Python, Perl
Logic Programming: ASP, CSP
Data Science: Numpy, Scikit-learn, Pandas, Seaborn, Tensorflow, Pytorch
Web Programming: HTML5, Javascript, CSS3, jQuery, Bootstrap4, JSP, Angular
Rest API: Django, Spring Boot
Multithread Programming: OpenMP, MPI
IGPE: Allegro, JavaFX, Java Swing
Relational Database
Data cleaning: Pentaho
Data visualization and data analysis: Tableau
Process mining: Disco, ProM

LANGUAGE SKILLS

Mother tongue(s): **ITALIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	B2	B2	B2	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user